

A cross-climatic analysis of building thermal performance in office buildings under climate change

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Abstract

Climate change presents significant challenges to building thermal performance across diverse global contexts. With operational stage exceeding 50 years, office buildings designed using historical climate conditions will operate in the future. This study evaluates future office building performance across eight ASHRAE climate zones, from hot-humid to subarctic regions. Using a ten-storey office building prototype, simulations were conducted under current weather conditions and future Shared Socioeconomic Pathway (SSP) scenarios for 2050 and 2080 via Ladybug Tools in Grasshopper. The results demonstrate substantial increases in cooling energy demand (35-106% under SSP585 by 2080) while exhibiting significant reductions in heating requirements (26-60% under SSP585 by 2080). Temperate regions display transitional behavior with relatively stable total energy consumption but fundamental shifts in seasonal heating-cooling ratios. Peak load analysis indicates critical implications for HVAC system sizing and grid infrastructure, with cooling peak loads increasing by up to 48.77% in certain regions while heating peaks decline by up to 39.06%. Thermal comfort metrics show improving conditions in colder climates but deteriorating comfort in warmer regions, necessitating regionally differentiated adaptation strategies. This cross-climate comparative analysis contributes to understanding long-term building performance under global warming.

Highlights

- Climate change increases cooling demand 35-106% while reducing heating 26-60% by SSP585_2080
- Peak cooling loads increase up to 48.77%; heating peaks decline 39.06%.
- Thermal comfort improves in cold regions but worsens in warm climates

Introduction

Human activities have significantly contributed to global warming, resulting in substantial temperature increases since pre-industrial times. According to the IPCC 6th report, human-induced climate change elevates the global average surface air temperature by 0.8-1.3°C from 2010-2019 compared to 1850-1900 (IPCC, 2021). The building sector represents a substantial contributor to this warming through both operational energy consumption and embodied emissions from construction materials.

Specifically, the building and construction sector contributes to 34% of total energy consumption and 37% of associated carbon emissions (UNEP, 2024). Recent data from the IEA indicates a 1% increase in building energy consumption in 2022 (IEA, 2024). These increasing trends pose significant challenges to achieve carbon neutrality by mid-century and limiting global temperature increase to 1.5°C above pre-industrial levels.

Climate change presents significant challenges for both human beings and the built environment. The manifestation of these challenges includes rising temperatures, fluctuating radiation patterns and humidity levels, and intensified extreme events such as heatwaves and cold snaps. Such changes significantly impact grid infrastructure and building operational resilience (Hong et al., 2023; Yan et al., 2025). Commercial office buildings that operate for more than 50 years must navigate these climate-related challenges, as they will all experience and operate under future climate conditions.

The relationship between global warming and building thermal performance has garnered significant research attention (Zeng et al., 2025; Yan et al., 2025; Shen et al., 2025), with increasing publications in recent years. This research primarily encompasses three key domains:

First, the development and implementation of future weather files/tools have emerged as a crucial research focus. For instance, Zeng et al. (2025) comprehensively reviewed downscaling methods, encompassing both statistical and dynamic approaches. Among these methodologies, morphing, proposed by Belcher et al. (2005), has emerged as the most popular method. This method employs three generic operations (shift, stretch, and combination) to modify climate variables. Considering the great efforts and time required in these processes, several specialized tools have been developed, including CCWorldWeatherGen, WeatherShift, Meteororm, and Future Weather Generator. Notably, Jentsch et al. (2013) first developed the CCWorldWeatherGen to generate future weather data for A2 scenarios based on HadCM3 model. The previous three tools utilized outdated model and are combined with commercial tools. Thereby, Rodrigues et al. (2023) introduced an open-access tool incorporating latest SSP scenarios for 2050 and 2080.

Second, extensive research has investigated future building energy performance across multiple metrics, including heating and cooling energy demand, peak loads, carbon emissions, and thermal comfort. In a comprehensive study, Ciancio et al. (2020) investigated future residential building energy performance across 19 European cities. They revealed a general decrease in heating demand coupled with substantial increases in cooling electricity consumption, particularly in Southern Europe. Their projections indicated that by 2080, summer cooling demands could surge by 272% in Rome, 142% in Palermo, and 238% in Granada. Similarly, Xiong et al. (2023) reported significant heating demand reductions in historically cool regions like Harbin, China, while observing substantial cooling increases in cooling-dominant regions like Guangzhou. Furthermore, Jalali et al. (2023) reported declining overall energy demand in New Zealand's cold regions (Zones 5 and 6) but increasing trends in warm regions (Zones 1 and 2). Rahif et al. (2022) projected that overheating risk for Belgian nearly zero-energy buildings (NZEB) could escalate by 528% while overcooling risk may decrease by 32% by 2090s. D'Agostino et al. (2022) discussed the energy performance and PV outputs for NZEB under a changing climate in eight regions.

Third, research has increasingly focused on adaptation and mitigation strategies through design interventions, retrofitting solutions, and optimization. Pajek and Košir (2021) explored different passive energy conservation measures including opaque thermal transmittance, natural ventilation, shape factor, etc. for residential buildings. They reported that these interventions can only partly counterbalance but difficult to neutralize the effect of global warming on building energy. Yan et al. (2022) conducted multi-objective optimization of thermal comfort, energy and daylight in Singapore residential buildings, identifying a trend toward enhanced envelope thermal insulation as an adaptation strategy for future Singapore residential buildings. Additionally, Bamdad (2023) demonstrated that cool roof implementations could achieve cooling demand reductions of 14-22% in tropical and subtropical regions under current and future weather scenarios. Specifically, this intervention may increase total energy use in current weather in cold and mild climates but may offer potential savings in future scenarios for regions like Canberra and Melbourne.

Despite these advances, significant knowledge gaps remain in understanding building performance under climate change. The long-term effects on energy demand, peak loads, and thermal comfort across different global climate zones remain poorly understood. This study addresses these gaps by evaluating the future building thermal performance of office buildings in different ASHRAE climate zones. The research models different representative office building prototypes across multiple SSP scenarios. The objective aims to quantify shifts in annual energy consumption, indoor thermal comfort (PMV/PPD) and peak heating/cooling demands in

different climates. By doing so, the findings from this study would inform building design practices, retrofit strategies, and policy development, ultimately contributing to a resilient built environment in a warming climate.

Materials and Methods

This section outlines the approach adopted to evaluate the thermal performance of office buildings under baseline and future weather files in diverse climate conditions.

Case study and construction sets

To assess thermal performance under climate change, a ten-storey office building prototype (Figure 1) was selected to represent typical CBD office buildings, which has already been utilized in previous studies (Daly et al., 2014; Bell et al., 2022). The prototype maintains consistent geometric parameters with a square footprint of 31.6m × 31.6m, yielding a total floor area of 9,985.6 m². The floor height is set at 3.6m, with 30% window-to-wall ratio. The HVAC system employs a VAV system with PFP boxes, with heating and cooling setpoints at 20°C and 24°C, respectively.

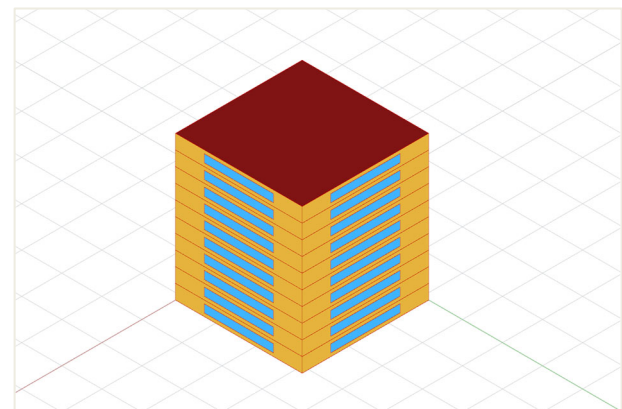


Figure 1: Ten-storey office building prototype.

Energy and thermal comfort simulations were executed using Ladybug Tools in Grasshopper. The building prototypes are differentiated through their construction sets, implemented using Honeybee construction sets aligned with ASHRAE climate zones. The 2019 vintage construction sets were utilized to represent enhanced thermal properties consistent with contemporary building standards. The construction sets incorporate climate-specific thermal envelope specifications designed to meet regional performance requirements while maintaining comparative consistency across zones. Wall assemblies feature U-values ranging from 0.46 W/(m²·K) in warmer climates (ASHRAE Zones 1-3) to 0.18 W/(m²·K) in colder regions (ASHRAE Zone 8). U-values of roof systems span from 0.15 W/(m²·K) to 0.09 W/(m²·K) across the climate zones. Window U-values varies from 2.86 W/(m²·K) in hot climates to 1.49 W/(m²·K) in subarctic areas. Solar heat gain coefficients (SHGC) are optimized regionally, ranging from 0.23 in cooling-dominated zones to 0.40 in heating-dominated climates to balance daylighting benefits with thermal performance.

By implementing zone-appropriate but technologically consistent construction standards, the study ensures that observed energy performance variations result primarily from climatic factors rather than disparities in building envelope efficiency. The simulation framework evaluates multiple metrics including PMV/PPD, annual heating and cooling demands (kWh/m²), peak loads (W/m²), and total energy use intensity (EUI).

Representative cities and future climate data

The study adopted the Future Weather Generator (Rodrigues, Fernandes, et al., 2023), a state-of-the-art tool developed for creating projected weather files under various climate scenarios. The Future Weather Generator processes current EnergyPlus Weather (EPW) files using climate change projections from the latest models such as EC-Earth3, specifically incorporating SSP scenarios. The tool generates hourly weather data for two future time periods: 2050 (2036–2065) and 2080 (2066–2095). Key weather parameters modified include dry-bulb temperature, humidity, global horizontal radiation. The baseline weather files are obtained from the ladybug EPW website. The SSP245 (intermediate scenario) and SSP585 (high emissions scenario) are considered for generating the future weather files (Chakraborty et al., 2021; Deng et al., 2023; Fereidani et al., 2023; Rodrigues, Fernandes, et al., 2023; Rodrigues, Fereidani, et al., 2023; Xiong et al., 2023; Yan et al., 2022).

To investigate the impact of global warming on building energy performance worldwide, the study encompasses multiple cities across eight ASHRAE climate zones (Table 1), ranging from very hot-humid (Zone 1) to subarctic conditions (Zone 8). This extensive geographical distribution enables robust assessment of climate change impacts across diverse baseline conditions. For each location, three temporal scenarios are analyzed: baseline climate, 2050s, and 2080s, with each future scenario incorporating both SSP245 and SSP585 projections.

Table 1: Selected representative cities in the different ASHRAE climate zones.

Zone	Description	Cities (Country)
1	Very hot	Miami (US)
2	Hot	Guangzhou (CN)
3	Warm	Melbourne (AU)
4	Mixed	London (UK)
5	Cool	Zürich (CH)
6	Cold	Helsinki (FI)
7	Very Cold	Edmonton (CA)
8	Subarctic	Sisimiut (GL)

Results and discussion

Annual energy performance

Figure 2 reveals distinct patterns in office energy consumption across ASHRAE climate zones, with annual HVAC EUI ranging from 116.94 kWh/m² in Melbourne

(ASHRAE 3) to 267.21 kWh/m² in Sisimiut (ASHRAE 8). Comparisons are made to validate the simulation results with previous studies that adopted the same building prototype. Specifically, Daly et al. (2014) reported total EUI values of approximately 150 kWh/m² for Melbourne, Canberra, and Sydney, which are considerably larger than our simulation results. These discrepancies can be attributed to several factors including internal heat gain assumptions (equipment and lighting efficiency), weather file variations, HVAC efficiency parameters, schedules, and building envelope thermal properties. In the Daly et al. (2014) study, lighting loads were set at 11 W/m². In contrast, this study assumes 4.5 W/m² for lighting in accordance with the NCC in Australia (Australian Building Codes Board, 2023). While these simulation-based comparisons provide valuable insights into energy performance trends, future research should incorporate field validation through real energy consumption data and occupant comfort surveys to enhance model credibility and robust cross-study comparisons. Buildings in Miami and Guangzhou demonstrate significant cooling energy consumption (91.81 and 79.95 kWh/m² respectively), which accounts for 49.17% and 44.82% of total EUI, respectively. Conversely, in subarctic regions in ASHRAE 8 such as Sisimiut, heating demand reaches 189.46 kWh/m², accounting for more than 70% of the total HVAC EUI. Interior lighting (17.79 kWh/m²) and electric equipment (47.54 kWh/m²) constitute a large share of total EUI, especially in mild temperature climates such as Melbourne, where heating and cooling contribute only 15.61 and 20.57 kWh/m² respectively. Regarding auxiliary energy systems, pump energy and heat rejection requirements demonstrate strong correlation with cooling demands, showing higher values in hot climates (ASHRAE 1&2).

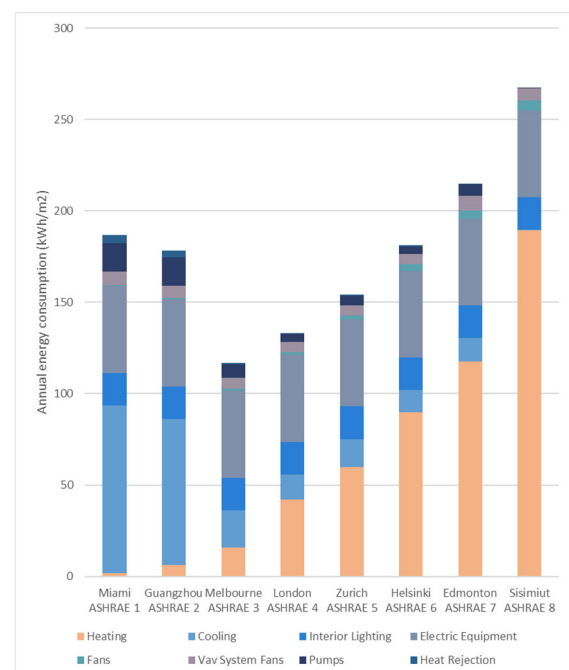


Figure 2: Annual energy consumption of office buildings in different ASHRAE climate zones.

Figure 3 illustrates the projected heating and cooling EUI trends across different climate zones and future slices, revealing shifts in building energy profiles under SSP scenarios. The analysis demonstrates a consistent pattern across all climate zones: decreasing heating energy demand and increasing cooling requirements, aligning with findings from previous research (Abolhassani et al., 2023; Wan et al., 2011). Specifically, heating energy consumption would decrease by 25.59-60.00% under SSP585_2080 scenarios, and 19.07-44.39% under SSP245_2080. Concurrently, cooling energy demand increases by 35.18-105.63% (excluding Sisimiut, which exhibits negligible cooling requirements) under SSP585_2080, and 24.99-64.01% under SSP245_2080.

In cold regions (Sisimiut, Edmonton and Helsinki), substantial reductions in heating energy consumption are projected. Specifically, Sisimiut demonstrates a 34.53% decrease (65.42 kWh/m² reduction) under SSP585 by 2080 compared to the baseline. For Helsinki, the projections indicate 54.26% reduction and 48.75 kWh/m² decrease for the same scenario. Conversely, in hot regions like Miami and Guangzhou in ASHRAE 1-2, cooling energy requirements are projected to increase substantially by 35.18% (32.3 kWh/m²) and 40.95% (32.74 kWh/m²) under SSP585_2080 compared to the baseline, respectively. Due to this significant cooling increases in tropical regions alongside substantial heating decreases in extreme cold regions, the heating and cooling energy consumption in Miami is expected to surpass that of Sisimiut (largest in the current situation), becoming the most energy-intensive location in the future SSP585_2080 scenario. This transition highlights the critical challenges associated with cooling-dominant cities under climate change. Moreover, these projections likely underestimate actual energy requirements, as they do not account for microclimate effects such as the urban heat island. As demonstrated by Akkose et al. (2021), climate change alone can increase annual cooling degree days from 119 to 335, while the combined effect with urban heat island further elevates this to 432 °C-day. Temperate climates such as Melbourne and London exhibit a pronounced shift toward cooling-dominant energy profiles. For example, in London, the heating to cooling ratio decreases from 3.04 in the baseline to 1.51 in SSP245_2080 and to 1.02 in SSP585_2080, indicating a fundamental transformation in seasonal energy demand patterns.

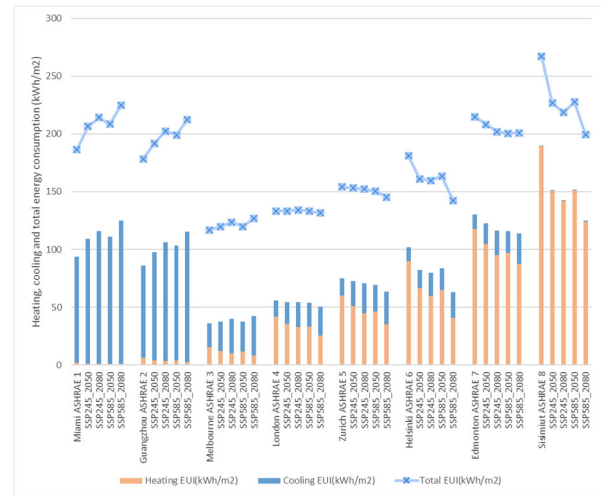


Figure 3: Future heating, cooling, and total energy consumption.

Regarding total HVAC energy consumption, the analysis reveals three distinct response patterns across the climate: climate change leads to increased total EUI in ASHRAE 1-3 regions, decreased total EUI in cold regions in ASHRAE 6-8, while temperate regions show relatively stable total EUI with internal shifts between heating and cooling components. This trend is consistent with findings from (Wang & Chen, 2014; Jalali et al., 2023). For instance, Jalali et al. (2023) reported decreasing thermal loads in cold zones 5&6 and increasing trends in warmer zones 1&2 in New Zealand. Similarly, Zou et al. (2021) identified three distinct trends in residential energy demand based on a study across 15 locations in China: decreasing consumption in zones (I, II, VI and VII), relatively small decreases in climate zones III and V in central China, and increasing energy requirements in zone IV where cooling is dominant.

Peak load assessment

Figure 4 illustrates the projected trends in peak loads of the office buildings across diverse ASHRAE climate zones under future climate scenarios, revealing significant implications for building systems design in a warming climate. Consistent with the patterns observed in annual energy consumption, the analysis demonstrates that heating peak loads will decrease whilst cooling peak loads will rise under all SSP scenarios. Notably, the high-emission scenarios (SSP585) produce significantly higher cooling demand compared to SSP245 pathways.



Figure 4: Building peak loads by location across climate scenarios

Buildings in hot-humid regions (ASHRAE 1–2) such as Miami and Guangzhou exhibit the most substantial increases in cooling peak loads. Specifically, the Miami cooling load rises from 46.72 kW (baseline) to 60.59 kW (SSP585_2080) (+29.69%), while heating demand declines by 39.06%. Similarly, Guangzhou experiences a cooling load increase from 47.81 kW to 64.04 kW (+33.95%), reflecting amplified heat stress in warm-temperate zones. These elevated peak loads could impose significant stress on electrical grid infrastructure and potentially lead to power outage (Hong et al., 2023). In addition, these projections do not incorporate extreme events like heatwaves and cold snaps, which are projected to intensify under climate change and could further exacerbate impacts on human health and grid infrastructure. Such substantial increases necessitate upgraded HVAC capacity, grid flexibility, and demand-side management to avoid overloads. Among the analyzed locations, buildings in Zurich (ASHRAE 5) demonstrate the greatest relative cooling peak load increase (Figure 5), with projections indicating gains of 25.78% under SSP245_2080 and 48.77% under SSP585_2080 scenarios.

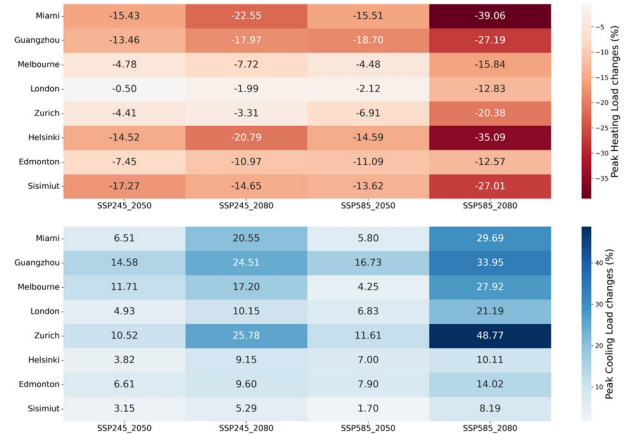


Figure 5: Percentage change heatmap for peak heating and cooling loads.

In contrast, cold-climate cities like Sisimiut and Edmonton experience sharp declines in heating loads. For example, heating peak load of office buildings in Edmonton would decrease from 216.54 kW to 189.33 kW under SSP585_2080 (−12.57%), and that of Sisimiut would reduce from 198.16 kW to 144.63 kW under SSP585_2080 (−27.01%). These buildings also exhibit slight increases in peak cooling loads of 14.02% and 8.19% under the same scenario, respectively. While reduced heating demand may lower operational heating energy consumption in colder regions, the transition toward electrification of space heating could still pose challenges for grid capacity during winter periods.

The analysis reveals distinct regional adaptation priorities: warmer zones (ASHRAE 1–2) face dominant cooling challenges, while extreme colder zones (ASHRAE 6–8) see heating-dominated systems become less critical. Based on these findings, targeted adaptation strategies should be implemented according to regional climate characteristics. Buildings in warm climates experiencing intensified heat stress should prioritize HVAC system upgrades, implementation of passive/active cooling strategies such as shading, and integration of renewable energy systems such as solar panels to offset increasing cooling loads. For buildings in cold and temperate climates, adaptation efforts should focus on retrofitting older building stock with adaptive insulation and ventilation systems capable of addressing both declining heating needs and increased cooling requirements.

Thermal comfort analysis

Figure 6 presents a comprehensive analysis of thermal comfort metrics, Predicted Mean Vote (PMV) and Percentage of People Dissatisfied (PPD), across the eight ASHRAE climate zones under current and SSP scenarios. The analysis reveals a warming pattern, with all office buildings across diverse climate zones showing increasing PMV values across temporal horizons and emission pathways, though with significant regional variations in magnitude.

In warm regions (ASHRAE zones 1-2), office buildings in Miami and Guangzhou during weekdays working hours already exhibit consistently positive PMV values (0.44 in the baseline increasing to 0.60 in SSP585_2080 for Miami). This progressive thermal intensification suggests that warmer climates will require substantial adjustments in building design strategies or HVAC systems to maintain acceptable thermal comfort under future climate conditions. Melbourne (Zone 3) represents a transition point with near-neutral thermal conditions at the baseline, projected to shift to slightly warm (PMV ~0.13) under SSP585_2080 scenarios. In cold climate zones (4-5), London and Zurich show slightly cool conditions (PMV -0.25 to -0.28) that trend toward neutral (-0.13 under SSP245_2080 and -0.07 under SSP585_2080) with climate change. This indicates potential comfort improvements in these office environments. In very cold, and subarctic climate zones (6-8), the analyzed office buildings exhibit the most negative PMV values, with Sisimiut (ASHRAE 8) showing the coldest sensation (PMV -0.82 in the baseline). While all scenarios show warming trends, these office buildings remain in the negative PMV range even under the most extreme scenario. These changes highlight the potential for a decrease in heating demand, although some discomfort due to cold conditions may still persist in colder regions.

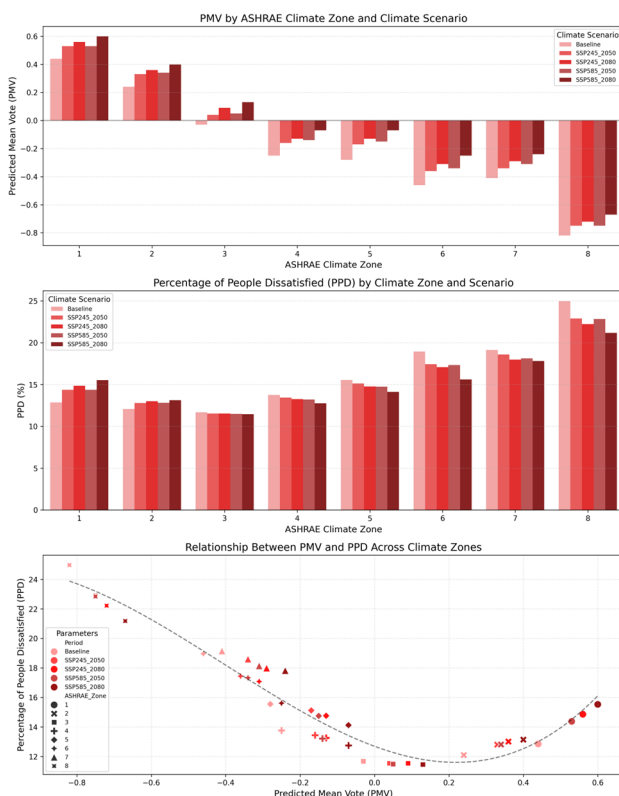


Figure 6: Thermal comfort analysis across different climate zones and scenarios.

Regarding occupant satisfaction metrics, the analysis of warm ASHRAE Climate Zones (1-2) indicates that Miami and Guangzhou show an increasing trend in PPD values, indicating deteriorating thermal comfort conditions in future office buildings. Under SSP585_2080 conditions,

Miami experiences an increase in PPD from 12.85% to 15.53%, while values for Guangzhou rise from 12.09% to 13.14% respectively. This progressive degradation in office comfort conditions as warming intensifies necessitates enhanced personal comfort measures and improved cooling strategies for maintaining acceptable thermal environments. Melbourne consistently maintains the lowest PPD levels (~11%) across all scenarios, suggesting it will remain the optimal comfort zone for office environments under various climate scenarios. For regions characterized by extreme cold conditions, such as Edmonton, Sisimiut, and Helsinki, the projections demonstrate significantly decreasing PPD values as the climate gets warmer. While climate warming is typically considered problematic, climate change actually improves thermal comfort in colder regions by reducing excessive cold sensations.

The scatter plot in the lower part of Figure 6 illustrates the strong inverse relationship between PMV and PPD across climate zones. As PMV deviates further from neutral (0) in either direction (colder/warmer conditions), PPD increases, showing greater dissatisfaction among occupants.

Furthermore, operational parameters significantly influence thermal comfort outcomes beyond climate effects alone. Specifically, heating and cooling setpoints would have substantial impacts on PMV and PPD metrics, with corresponding implications for energy consumption and occupant thermal comfort. Several studies have specifically investigated operational strategies for climate adaptation. Jafarpur and Berardi (2021) evaluated the impact of different setpoint temperatures on future energy demand and comfort, reporting potential building energy savings of 0.9-9.9% in three Canadian cities through setpoint adjustments. Tsoka et al. (2021) also reported that lowering the heating setpoint by 1°C led to 8.5-14.3% energy savings for future weather files. These findings align with research by Wan et al. (2012), who highlighted the significant energy saving potential of increased cooling setpoints. Such operational adjustments represent cost-effective adaptation strategies, particularly within the framework of adaptive thermal comfort models. Jafarpur and Berardi (2021) also reported that increasing the cooling setpoint by 1°C from 24°C to 25°C resulted in the percentages of zones being uncomfortable increasing by 24-51% for Canadian cities, while decreasing the heating setpoint from 21°C to 20°C increased the uncomfortable percentage by 27-67%. Furthermore, occupant behavioral adaptations such as clothing adjustments will influence thermal comfort outcomes, as individuals in different climates develop their own clothing preferences to adapt to environmental conditions. However, implications of such behavioral factors are beyond the research scope of this study.

Conclusion

This study explored the impact of climate change on office building thermal performance under diverse climate zones. The cross-climate analysis demonstrates that while all regions face shifting thermal demands, the magnitude of these changes varies substantially across the global climate regions. The results of this study can provide insights for building design, operation, and policy in a warming climate.

The significant increase in cooling energy demand and peak loads in warm climate zones (ASHRAE 1-2) indicates critical adaptation challenges that will require enhanced cooling capacity, active and passive design strategies, and potentially, renewable energy integration to mitigate grid impacts. Second, the transition of temperate climates toward cooling-dominated energy profiles necessitates flexible building energy conservation measures capable of adapting to shifting seasonal demands. Third, the improvement in thermal comfort alongside reduced energy demands in extreme cold regions represents a potential adaptation opportunity.

This research also has important implications for building professionals and policymakers. In warm climate regions, adaptation efforts should prioritize cooling efficiency, peak load management, and passive design strategies to maintain comfort while minimizing energy impacts. In temperate regions, building systems should emphasize flexibility to accommodate shifting seasonal demands. In cold regions, building codes and standards may need adjustment to recognize reduced heating requirements while preparing for emerging cooling needs.

These findings contribute to the growing body of research on climate-resilient buildings by providing a cross-comparison across diverse climate contexts and multiple thermal performance metrics. The methodology applied across climate zones enables direct comparison of office buildings regarding climate change vulnerability and adaptation potential globally. However, this research also suffers from certain limitations. The adoption of ASHRAE-based construction sets in Ladybug tools derived primarily from North American standards potentially limits the direct applicability of findings to building stocks in other geographical contexts with different construction traditions. Second, a significant limitation lies in the exclusion of urban microclimate effects and extremes. Such exclusion may underestimate energy requirements, peak loads for both heating and cooling and thermal comfort degradation.

References

Abolhassani, S.S., Mastani Joybari, M., Hosseini, M., Parsaee, M. & Eicker, U. (2023) A systematic methodological framework to study climate change impacts on heating and cooling demands of buildings. *Journal of Building Engineering*, **63**: 105428.

- Akkose, G., Meral Akgul, C. & Dino, I.G. (2021) Educational building retrofit under climate change and urban heat island effect. *Journal of Building Engineering*, **40**: 102294.
- Australian Building Codes Board (2023) *National Construction Code* 2022. <https://ncc.abcb.gov.au/>, accessed 7 January 2025.
- Bamdad, K. (2023) Cool roofs: A climate change mitigation and adaptation strategy for residential buildings. *Building and Environment*, **236**: 110271.
- Belcher, S., Hacker, J. & Powell, D. (2005) Constructing design weather data for future climates. *Building Services Engineering Research and Technology*, **26** (1): 49–61.
- Bell, N.O., Bilbao, J.I., Kay, M. & Sproul, A.B. (2022) Future climate scenarios and their impact on heating, ventilation and air-conditioning system design and performance for commercial buildings for 2050. *Renewable and Sustainable Energy Reviews*, **162**: 112363.
- Chakraborty, D., Alam, A., Chaudhuri, S., Başağaoğlu, H., Sulbaran, T. & Langar, S. (2021) Scenario-based prediction of climate change impacts on building cooling energy consumption with explainable artificial intelligence. *Applied Energy*, **291**: 116807.
- Ciancio, V., Salata, F., Falasca, S., Curci, G., Golasi, I. & de Wilde, P. (2020) Energy demands of buildings in the framework of climate change: An investigation across Europe. *Sustainable Cities and Society*, **60**: 102213.
- D'Agostino, D., Parker, D., Epifani, I., Crawley, D. & Lawrie, L. (2022) How will future climate impact the design and performance of nearly zero energy buildings (NZEBs)? *Energy*, **240**: 122479.
- Daly, D., Cooper, P. & Ma, Z. (2014) Implications of global warming for commercial building retrofitting in Australian cities. *Building and Environment*, **74**: 86–95.
- Deng, Z., Javanroodi, K., Nik, V.M. & Chen, Y. (2023) Using urban building energy modeling to quantify the energy performance of residential buildings under climate change. *Building Simulation*. <https://doi.org/10.1007/s12273-023-1032-2>, accessed 17 August 2023.
- Fereidani, N.A., Rodrigues, E. & Gaspar, A.R. (2023) The effectiveness of the Iranian building code in mitigating climate change in Bandar Abbas. *Energy for Sustainable Development*, **76**: 101266.
- Hong, T., Malik, J., Krelling, A., O'Brien, W., Sun, K., Lamberts, R. & Wei, M. (2023) Ten questions concerning thermal resilience of buildings and occupants for climate adaptation. *Building and Environment*: 110806.

- IEA (2024) Buildings - Energy System [Internet]. <https://www.iea.org/energy-system/buildings>, accessed 3 January 2025.
- IPCC (2021) *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*.
- Jafarpur, P. & Berardi, U. (2021) Effects of climate changes on building energy demand and thermal comfort in Canadian office buildings adopting different temperature setpoints. *Journal of Building Engineering*, **42**: 102725.
- Jalali, Z., Shamseldin, A.Y. & Ghaffarianhoseini, A. (2023) Impact assessment of climate change on energy performance and thermal load of residential buildings in New Zealand. *Building and Environment*, **243**: 110627.
- Jentsch, M.F., James, P.A.B., Bourikas, L. & Bahaj, A.S. (2013) Transforming existing weather data for worldwide locations to enable energy and building performance simulation under future climates. *Renewable Energy*, **55**: 514–524.
- Pajek, L. & Košir, M. (2021) Strategy for achieving long-term energy efficiency of European single-family buildings through passive climate adaptation. *Applied Energy*, **297**: 117116.
- Rahif, R., Norouzasas, A., Elnagar, E., Doutreloup, S., Pourkiaei, S.M., Amaripadath, D., Romain, A.-C., Fettweis, X. & Attia, S. (2022) Impact of climate change on nearly zero-energy dwelling in temperate climate: Time-integrated discomfort, HVAC energy performance, and GHG emissions. *Building and Environment*, **223**: 109397.
- Rodrigues, E., Fereidani, N.A., Fernandes, M.S. & Gaspar, A.R. (2023) Climate change and ideal thermal transmittance of residential buildings in Iran. *Journal of Building Engineering*, **74**: 106919.
- Rodrigues, E., Fernandes, M.S. & Carvalho, D. (2023) Future weather generator for building performance research: An open-source morphing tool and an application. *Building and Environment*, **233**: 110104.
- Shen, P., Li, Y., Gao, X., Zheng, Y., Huang, P., Lu, A., Gu, W. & Chen, S. (2025) Recent progress in building energy retrofit analysis under changing future climate: A review. *Applied Energy*, **383**: 125441.
- Tsoka, S., Velikou, K., Tolika, K. & Tsikaloudaki, A. (2021) Evaluating the Combined Effect of Climate Change and Urban Microclimate on Buildings' Heating and Cooling Energy Demand in a Mediterranean City. *Energies*, **14** (18): 5799.
- UNEP (2024) *2023 Global Status Report for Buildings and Construction: Beyond foundations - Mainstreaming sustainable solutions to cut emissions from the buildings sector*. United Nations Environment Programme. <https://wedocs.unep.org/20.500.11822/45095>, accessed 2 May 2024.
- Wan, K.K.W., Li, D.H.W., Liu, D. & Lam, J.C. (2011) Future trends of building heating and cooling loads and energy consumption in different climates. *Building and Environment*, **46** (1): 223–234.
- Wan, K.K.W., Li, D.H.W., Pan, W. & Lam, J.C. (2012) Impact of climate change on building energy use in different climate zones and mitigation and adaptation implications. *Applied Energy*, **97**: 274–282.
- Wang, H. & Chen, Q. (2014) Impact of climate change heating and cooling energy use in buildings in the United States. *Energy and Buildings*, **82**: 428–436.
- Xiong, J., Guo, S., Wu, Y., Yan, D., Xiao, C. & Lu, X. (2023) Predicting the response of heating and cooling demands of residential buildings with various thermal performances in China to climate change. *Energy*, **269**: 126789.
- Yan, D., Wu, Y., Malik, J. & Hong, T. (2025) Ten questions on future and extreme weather data for building simulation and analysis in a changing climate. *Building and Environment*, **269**: 112461.
- Yan, H., Ji, G. & Yan, K. (2022) Data-driven prediction and optimization of residential building performance in Singapore considering the impact of climate change. *Building and Environment*, **226**: 109735.
- Zeng, Z., Kim, J.-H. (Jeannie), Tan, H., Hu, Y., Cameron-Rastogi, P., Villa, D., New, J., Wang, J. & Muehleisen, R.T. (2025) A review of future weather data for assessing climate change impacts on buildings and energy systems. *Renewable and Sustainable Energy Reviews*, **212**: 115213.
- Zou, Y., Xiang, K., Zhan, Q. & Li, Z. (2021) A simulation-based method to predict the life cycle energy performance of residential buildings in different climate zones of China. *Building and Environment*, **193**: 107663.